Memory Hierarchy

CISC 221 – Assignment 5

1. (2 marks; note: 1MB = 1024×1024B)

6.24 ◆◆

Suppose that a 2 MB file consisting of 512-byte logical blocks is stored on a disk drive with the following characteristics:

Parameter	Value
Rotational rate	18,000 RPM
$T_{\text{avg seek}}$	8 ms
Average number of sectors/track	2,000
Surfaces	4
Sector size	512 bytes

For each case below, suppose that a program reads the logical blocks of the file sequentially, one after the other, and that the time to position the head over the first block is $T_{\text{avg seek}} + T_{\text{avg rotation}}$.

A. *Best case:* Estimate the optimal time (in ms) required to read the file given the best possible mapping of logical blocks to disk sectors (i.e., sequential).

B. *Random case:* Estimate the time (in ms) required to read the file if blocks are mapped randomly to disk sectors.

2. (4 marks)

6.25 ◆

The following table gives the parameters for a number of different caches. For each cache, fill in the missing fields in the table. Recall that m is the number of physical address bits, C is the cache size (number of data bytes), B is the block size in bytes, E is the associativity, S is the number of cache sets, t is the number of tag bits, s is the number of set index bits, and b is the number of block offset bits.

Cache	m	C	B	E	S	t	s	b
1.	32	1,024	4	4	_____	_____	_____	_____
2.	32	1,024	4	256	_____	_____	_____	_____
3.	32	1,024	8	1	_____	_____	_____	_____
4.	32	1,024	8	128	_____	_____	_____	_____
5.	32	1,024	32	1	_____	_____	_____	_____
6.	32	1,024	32	4	_____	_____	_____	_____

3. (2 marks)

2-way set associative cache

	Line 0						Line 1					
Set index	Tag	Valid	Byte 0	Byte 1	Byte 2	Byte 3	Tag	Valid	Byte 0	Byte 1	Byte 2	Byte 3
0	09	1	86	30	3F	10	00	0	—	—	—	—
1	45	1	60	4F	E0	23	38	1	00	BC	0B	37
2	EB	0	—	—	—	—	0B	0	—	—	—	—
3	06	0	—	—	—	—	32	1	12	08	7B	AD
4	C7	1	06	78	07	C5	05	1	40	67	C2	3B
5	71	1	0B	DE	18	4B	6E	0	—	—	—	—
6	91	1	A0	B7	26	2D	F0	0	—	—	—	—
7	46	0	—	—	—	—	DE	1	12	C0	88	37

For the cache memory shown above

- The memory is byte addressable.
- Memory accesses are to **1-byte words** (not to 4-byte words).
- Addresses are 13 bits wide.
- The cache is two-way set associative ($E = 2$), with a 4-byte block size ($B = 4$) and eight sets ($S = 8$).

A- List all of the hex memory address that will hit in set 1
B- List all of the hex memory addresses that will hit in set 6

Deliverables

To OnQ:

1. Single PDF file for all questions.